

Stage 4.3 - Final Project Summary

Conservative summary of the completed initial screen and full-discovery bidirectional extension

25 September 2026

This summary is intentionally evidence-calibrated. 'Retained' refers to the frozen descriptive screen and is not a statistical-significance or circuit-membership claim.

1. Status

Stage 4.3 is complete under the bounded plan. The initial 40-pair noising screen evaluated 305 configurations and selected 16 route contrasts plus 11 direct prerequisites. The final extension then evaluated exactly those 27 configurations on all 178 discovery pairs in both noising and reverse restoration, producing 9,612 records. Held-out families remained sealed.

Item	Result
Initial screen	40 pairs, 20 families, 12,200 noising records
Extension	178 pairs, 89 families, 9,612 bidirectional records
Selected contrasts	16 (frozen before full extension)
Retained in full extension	16/16 noise; 16/16 restore
Coherent in full extension	14/16 noise; 14/16 restore
Heterogeneous in full extension	16/16 noise; 16/16 restore
Held-out	sealed

2. Main results

Tested dependency	Final observation	Conservative interpretation
L17H1 -> L18 pair -> later Q	Largest and most consistent composed effects. Joint pair -> L27H6 Q: +1.635 noise / +1.511 restore; 98.9% family directional sign match.	Supports the tested ordered dependency; does not establish unique/minimal or literal semantic coding.
L15H25 -> L16 pair -> L18 Q	Three selected chains remain coherent. Means range about +0.151 to +0.222 across directions; 96.6% family directional sign match.	Supports composition under the joint selective-live-head intervention; individual necessity is not established.
L8H15 -> MLP9-associated -> L18 V	Four selected routes remain coherently negative, roughly -0.131 to -0.160 noise and -0.133 to -0.154 restore; 97.8-100% family sign match.	Opposing/attenuating intervention dependency, not 'negative information'.
L26H23 full downstream release	-0.183 noise / -0.166 restore; coherent in both; 91.0% family sign match.	Block-mediated evidence only.
L26H23 block27 only	-0.089 / -0.084; retained only as heterogeneous; 86.5% family sign match.	Should not be called coherent at full-discovery scale.
L20H7 full downstream release	-0.069 / -0.061; heterogeneous-only; 47.2% family sign match; family correlation -0.527.	Cancellation-prone / directionally unstable; not a stable directed route.

3. What changed relative to the 20-family screen

- All 16 selected routes keep the same aggregate mean sign when expanded to 89 discovery families and are retained in both directions.
- The major L17H1, L15H25, and L8H15 results remain qualitatively stable.
- L26H23 block27-only drops below the coherent signed-mean threshold and is heterogeneous-only.
- L20H7 drops more clearly below the coherent threshold and shows poor family-level agreement between noising and restoration.

- The 20-family subset is reproduced to floating-point precision inside the extension pipeline, supporting implementation consistency but not constituting independent validation.
- A secondary analysis of the 69 discovery families not used for selection gives the same 14-coherent / 2-heterogeneous-only classification. These 69 families are selection-excluded discovery data, not held-out data.

4. Negative and scope-limited findings

- No L13H18 or L16H31 route passed the initial frozen screen, so neither source entered the 178-pair extension. S4.3 therefore provides no new full-discovery claim for them.
- Six initially retained routes were omitted solely by the fixed 16-contrast cap and remain 20-family-only findings.
- No broad second search, within-block head search, or automatic fallback scan was run.
- Family averaging does not imply prompt-order invariance; the largest L17H1 -> L27H6 chain changes substantially in magnitude across the two orders.

5. Compatibility and provenance

The two family-034 orders exceeded the original historical-margin compatibility tolerance in a deterministic way on both A100-40GB and A100-80GB. A provenance audit verified that the authenticated Stage-3 prompts and historical logits were preserved exactly in the frozen S4.3 inputs. The final runner therefore kept the 0.05 threshold for the other 176 pairs and required exactly those two family-034 pairs to reproduce a frozen baseline diagnostic reference within 1e-6. They did so with zero reference error and remained in the scientific population.

6. Final claim boundary

S4.3 supports several route-specific and composed causal dependencies under the exact hybrid interventions, but it does not establish a complete semantic circuit, unique/minimal mediation, literal semantic codes, additive information shares, or held-out generalization. The strongest language should be reserved for consistency of the measured intervention effects, not for claims about what the model 'represents' internally.

7. Next step

Treat S4.3 as frozen and complete. If the project proceeds to held-out validation or to S4.4/S4.5 group/mechanism tests, freeze the route list, signs, thresholds, and interpretation before accessing held-out families. Do not use held-out data to select new S4.3 routes.

Reproducibility

Production identity: 1f6d18be982d12ce7d1c0a4a88f19f2fe94bff4c5af89ee66c6bb46a2836039b

Post-extension analysis: S4.3_post_extension_full89_bidirectional_v1

Analysis manifest SHA-256: 7833704dcda0ff85e78b58bbcd7a7eac1211f6810fa15ae588204ea230d33007